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PRRSV2 genetic diversity defined by RFLP patterns in the United States from 2007 to 2019

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Abstract. The genetic diversity of porcine reproductive and respiratory syndrome virus (PRRSV) increases over time. In 1998, restriction-fragment length polymorphism (RFLP) pattern analysis was introduced to differentiate PRRSV wild-type strains from VR2332, a reference strain from which a commercial vaccine (Ingelvac PRRS MLV) was derived. We have characterized here the PRRSV genetic diversity within selected RFLP families over time and U.S. geographic space, using available ISU-VDL data from 2007 to 2019. The 40,454 ORF5 sequences recovered corresponded to 228 distinct RFLPs. Four RFLPs [2-5-2 (21.2%), 1-7-4 (15.6%), 1-4-4 (11.8%), and 1-8-4 (9.9%)] represented 58.5% of all ORF5 sequences and were used for cluster analysis. Over time, there was increased detection of RFLPs 2-5-2, 1-7-4, 1-3-4, 1-3-2, and 1-12-4; decreased detection of 1-4-2, 1-18-4, 1-18-2, and 1-2-2; and different detection trends for 1-8-4, 1-4-4, 1-26-1, 1-22-2, and 1-2-4. An over-time cluster analysis revealed a single cluster for RFLP 2-5-2, supporting that sequences within RFLP 2-5-2 are still relatively conserved. For 1-7-4, 1-4-4, and 1-8-4, there were multiple clusters. State-wise cluster analysis demonstrated 4 main clusters for RFLP 1-7-4 and 1-8-4, and 6 for RFLP 1-4-4. For the other RFLPs, there was a significant genetic difference within them, particularly between states. RFLP typing is limited in its ability to discriminate among different strains of PRRSV. Understanding the magnitude of genetic divergence within RFLPs helps develop PRRSV regional control programs, placement, herd immunization strategies, and design of appropriate animal movements across borders to minimize the risk of PRRSV transmission.

Keywords: ORF5; RFLP; pairwise distance; phylogenetic; PRRSV; swine.

Porcine reproductive and respiratory syndrome (PRRS) is a devastating disease that affects swine and causes productivity losses. PRRS was initially described as mystery swine disease,[10,18,65] and first identified in the early 1990s in Europe[57] and the United States.[2,5] PRRS virus (PRRSV) is a RNA virus[69] that, during the last 30 y, has spread to most pig-producing countries. PRRSV is one of the most economically significant pathogens affecting the global swine population, causing economic losses estimated between €3 and €160 per sow in Europe,[36] and $560 million yearly in the U.S. swine herd.[11,12]

PRRSV is classified into 2 distinct species: PRRSV1 (*Betaarterivirus suid 1*) and PRRSV2 (*Betaarterivirus suid 2*).[20,67] These 2 genetically distinct species are distributed worldwide. PRRSV1, also known as the European type, has the Lelystad isolate as the prototype strain,[2,57] is predominant in Europe, but is also detected in the Americas[60] and Asia. PRRSV2, also known as the North American type, has ATCC VR-2332 as the prototype strain,[2,65] and is predominant in the Americas and Asia.[3,32,35,47–50,60,69] The 2 species of PRRSV have ~44% nucleotide difference.[20,69]

The whole genome of PRRSV is ~15.1 kb in length[20,28,35] and is organized into at least 10 open reading frames (ORFs).[3] Within the PRRSV genome, the ORF5 region has the highest genetic diversity[19] and thus is extensively used in molecular epidemiology and evolutionary studies. ORF5 is composed of 603–606 bp of nucleotides representing ~4% of the PRRSV genome.[3,69]

A method to differentiate PRRSV2 isolates was developed during the late 1990s and relied on restriction-fragment length polymorphism (RFLP) characteristics.[66] RFLP typing consists of assigning 3 numeric codes according to the patterns of digestion by the enzymes *Mlu*I, *Hinc*II, and *Sac*II on restriction sites within ORF5.[66] This method was developed primarily to differentiate strains closely related to the attenuated PRRSV Ingelvac modified-live vaccine (MLV; RFLP 2-5-2,

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Boehringer Ingelheim) from wild-type PRRSV strains, and it was easily understood by veterinarians and producers.[33,66] The sites targeted by the 3 enzymes can be predicted by software programs when there is information on the ORF5 sequence. Using ORF5 sequence data, veterinary diagnostic laboratories (VDLs) run an algorithm to establish the predicted RFLP pattern and report it to clients. In addition to grouping ORF5 sequences by the RFLP pattern, a method has been proposed based on lineages to group ORF5 sequences sharing similarities.[3,22,37,55]

RFLP pattern information has been used to associate specific virus strains with clinical outcomes of PRRSV infection, including morbidity and mortality,[7,8,25,29,31] and abortions.[1] Eventually, the RFLP pattern was also used to describe the virulence of viral strains.[8] Moreover, RFLP pattern information has been plugged into statistical models as an explanatory variable to measure the economic losses associated with specific wild-type strains.[24,25]

Given the ever-expanding genetic variability of PRRSV, the accuracy of the RFLP genotyping method to continue to differentiate RFLP 2-5-2 (Ingelvac MLV–like) from other patterns, or to differentiate between wild-type PRRSV strains,[62] has become questionable. However, no alternative genetic typing methods have been described to differentiate isolates.[62] As a consequence, RFLP typing remains in use by the swine industry and is used routinely to compare strains. Therefore, there is the need to understand the frequency of detection of RFLP patterns, and the contemporary genetic variation within RFLP patterns over time and geographic space. This information helps veterinarians, producers, and researchers make informed use of the RFLP pattern data.

Our objective was to characterize PRRSV genetic diversity within selected RFLP families over time and geographic space using available data from the Iowa State University Veterinary Diagnostic Laboratory (ISU-VDL; Ames, IA, USA). The specific aims included characterization of the PRRSV genetic diversity over time and between regions, and development of the capability to monitor the frequency of RFLP pattern detection over time.

Materials and methods

Veterinary diagnostic data containing submission information for accession ID, animal ID, received date, SNOMED clinical terms (CT; i.e., specimen), farm type, age, reason for submission, site state, and the respective PRRSV ORF5 sequences were gathered and retrieved from the ISU-VDL laboratory information management system (LIMS) comprising the period 01/01/2007 to 12/31/2019 (Fig. 1). Also, cluster analysis and pairwise comparisons to the most frequently detected RFLP patterns revealed a description of the PRRSV ORF5 diversity over time and across U.S. states.

Sequencing

PRRSV ORF5 sequencing were performed using a Sanger[52] technique at the ISU-VDL.[68] In brief, the Sanger technique targets specific regions of the PRRSV RNA, allowing recovery of specific consensus segments of the genetic sequence,[52] typically 603 to 606 nucleotides. Recovered ORF5 sequences were used to predict the RFLP pattern by comparing ORF5 sequences with the RFLP pattern bank information maintained by the University of Minnesota (Rossow S, pers. comm., 2020).

Data management and analysis

All recovered data were uploaded to SAS v.9.4 (SAS Institute) to prepare the data for cluster analysis. A unique identifier was assigned to each ORF5 sequence. An algorithm using the commands DATA STEP, PROC SQL, and PROC SORT was written to standardize the information for specimen based on the field "SNOMED CT". ORF5 sequences were aligned and compared within and between years and states. Additionally, ORF5 sequences were compared to the vaccine strains: Ingelvac MLV (Ingelvac PRRS MLV, Boehringer Ingelheim; RFLP 2-5-2), Ingelvac ATP (Ingelvac PRRS ATP, Boehringer Ingelheim; RFLP 1-4-2), Fostera (Fostera PRRS, Zoetis; RFLP 1-3-2), Primepac (Prime Pac PRRS+, Merck; RFLP 1-4-4), and Prevacent (Prevacent PRRS, Elanco; RFLP 1-8-4). Following the standard procedure adopted at the ISU, when an ORF5 sequence had ≥99% similarity with a vaccine strain, it was reported as "vaccine-like"; otherwise, it was reported as "wild-type" under a field named "PRRSV-similarity."

Data exclusion

Data from samples that originated from a non-U.S. territory based on the field "site state" were removed. After performing a pairwise comparison, PRRSV1 ORF5 sequences, identified by having ≥80% similarity with the Lelystad[28] strain were removed from analysis. For the remaining sequences, only those that had >60% similarity to at least one of the PRRSV2 comparison strains (i.e., VR2332 and vaccine strains) were retained in the analysis. After submitting PRRSV2 ORF5 sequences for RFLP prediction, the ones that resulted in an incomplete predictable RFLP pattern based on missing predicted enzymes *Mlu*I, *Hinc*II, and *Sac*II were removed. As an example, if a RFLP was precited as 1-?-4, the corresponding sequence data were removed (Fig. 1).

Based on the predicted RFLP pattern, only sequences that belonged to RFLP patterns that cumulatively represented ≥50% of all PRRSV2 ORF5 sequences were included in the pairwise and cluster analyses over time and states. In other words, sequences from low-prevalence PRRSV2 RFLP patterns were not included in the cluster analyses.

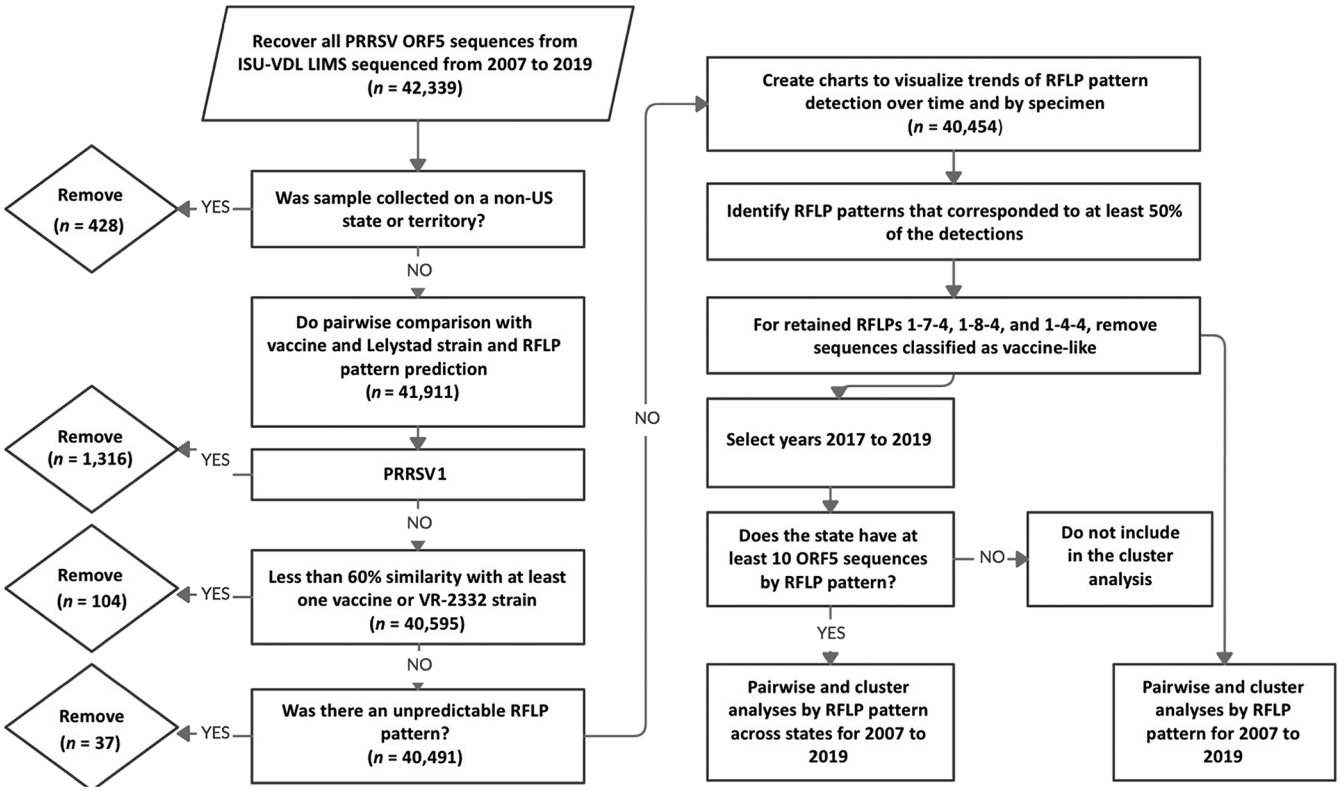

Figure 1. Flowchart for recovery and analyses of open reading frame 5 (ORF5) sequences recovered from the Iowa State University Veterinary Diagnostic Laboratory (ISU-VDL) laboratory information management system (LIMS). The numbers in parentheses are the numbers of sequences included in the analysis.

For RFLP 2-5-2, all ORF5 sequences, either classified as vaccine-like or wild-type, were retained for pairwise and cluster analysis. In contrast, for all other RFLP patterns, the ORF5 sequences classified as vaccine-like strains were removed and only non–vaccine-like ORF5 sequences were included for the pairwise and cluster analyses. As an example, sequences classified as Prevacent-like were removed from the RFLP 1-8-4 analyses but non–Prevacent-like strains (<99% ORF5 nucleotide identity to Prevacent vaccine strain) with RFLP 1-8-4 were included for RFLP 1-8-4 analysis.

To protect VDL clientele confidentiality, only states having at least 10 ORF5 sequences during 2017–2019 were included for pairwise and cluster analyses.

Cluster analyses

Grouping of PRRSV ORF5 sequences having similar genetic proximity can be achieved by applying cluster analysis. To perform cluster analysis on available ORF5 sequences, we used the discriminant analysis of principal component (DAPC) method.[16] DAPC analysis summarizes the differences between clusters while neglecting within-cluster variation.[16] DAPC analysis allows visualization, in scatterplots, of accumulated genetic differences between clusters. Individuals are represented by dots; groups sharing genetic similarities are represented by different colors and by an ellipse. The separation between groups is represented by accumulation of genetic differences in the DAPC's first and second principal components. The primordial level of separation between groups is represented by the first component and is represented in the scatterplot by the horizontal line. The second level of separation between groups is represented by the second component and is represented in the scatterplot by the vertical line. The interception point between the vertical and horizontal line indicates no genetic separation across groups.

Two levels of cluster analysis were performed. The first explored the genetic diversity of ORF5 over time and used sequences from 2007 to 2019. The second explored the genetic diversity across geographic regions, represented by U.S. states, and only used ORF5 sequences recovered during the 3 most recent years (2017–2019). The ORF5 sequences were aligned using the default setting of MAFFT software v.7.419.[17] Thereafter, the aligned sequences were imported into R[45] using the ape package.[38] Then, the presence of clusters was assessed by DAPC[16] using the adegenet package.[15]

Pairwise analysis

Pairwise distances were assessed for the same groups of sequences included in the cluster analysis. Pairwise distances were assessed over time, comprising 2007–2019. In an effort

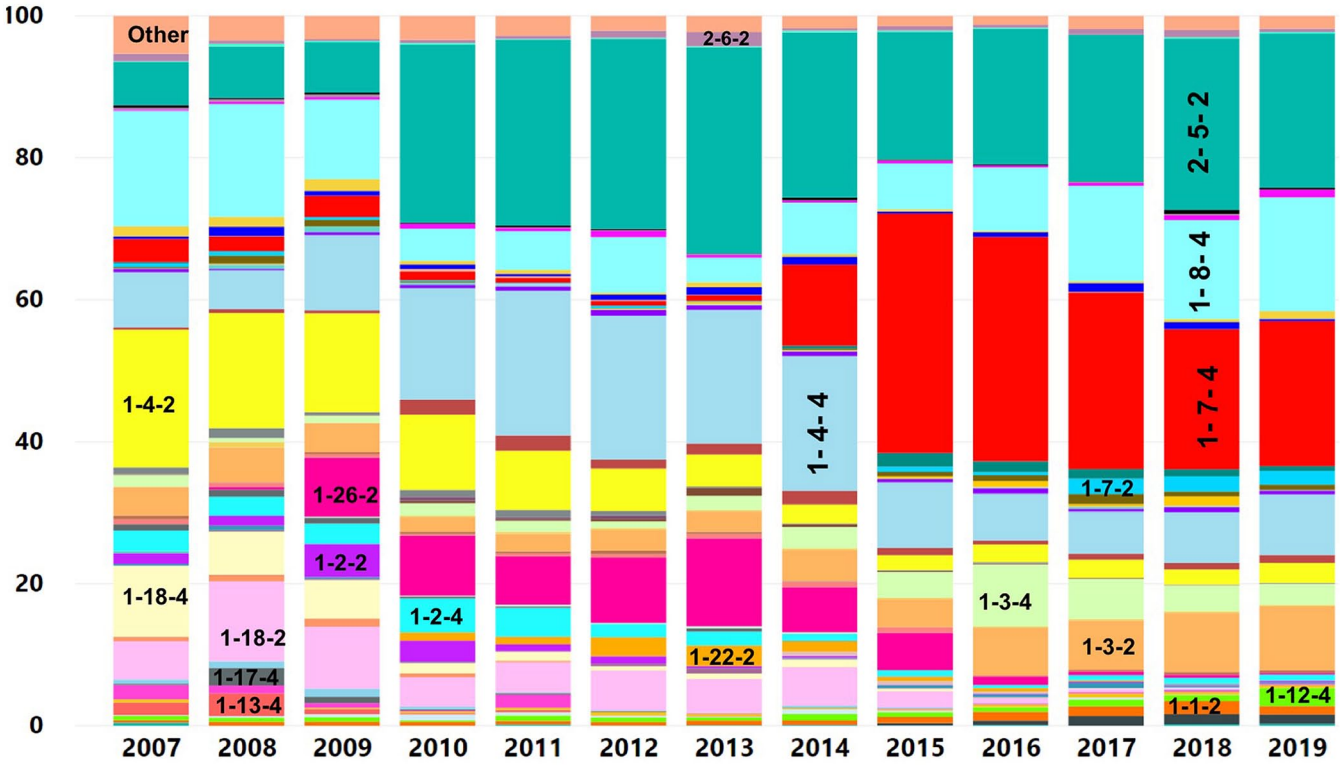

Figure 2. Trends of restriction-fragment length polymorphism (RFLP) pattern detection over time at the Iowa State University Veterinary Diagnostic Laboratory. Each color represents a different RFLP pattern, organized in ascending order for *Mlu*I, *Hinc*II, and *Sac*II enzyme patterns. Vertical labels represent the 4 most frequently detected RFLP patterns. Horizontal labels are additional RFLP patterns. The x-axis labels are the years of detection from 2007 to 2019. The y-axis represents the percentage of detections within each year.

to investigate the most current genetic diversity across states, only ORF5 sequences collected during 2017–2019 were retained. Pairwise distances and standard error were calculated using MEGA X,[21] changing from the default setting procedures to the bootstrap method. Three levels of information were generated: overall, within groups, and between groups.

Data visualization

Information for the received date, RFLP pattern, specimens, and PRRSV-similarity (i.e., vaccine-like or wild-type) were transferred to a data visualization tool (Power BI; Microsoft), allowing the visualization of changes in trends of RFLP pattern detection over time and to reveal the most frequently detected RFLP patterns (Fig. 1). For visualization purposes, all RFLP patterns that had up to 30 detections within the summarized period were grouped as "other".

Comparison of RFLP patterns within and between PRRSV lineages

PRRSV lineages were reported for the RFLPs detected most frequently in the database.[55] For this step, the reference ORF5 sequences for each lineage[37,55] were downloaded from GenBank and added to our sequence library. Sequences were aligned in MAFFT v.7.419 with the default setting. Neighbor-joining tree estimation was performed using the Phangorn package in R.[54] Rooted trees for each RFLP pattern comparison, considering the sequence ATCC VR-2332 (GenBank U87392) as the reference, were built and visualized using Dendroscope.[13,14]

Results

Visualizing RFLP patterns and identification of trends

A dataset of 42,339 ORF5 sequences was retrieved from the ISU-VDL LIMS for 2007–2019. A total of 428 were from non-U.S. states, 1,316 were PRRSV1, 104 had <60% similarity with compared strains, and 37 sequences had an incomplete predicted RFLP pattern; these sequences were removed from the database, leaving 40,454 ORF5 sequences (Fig. 1). A total of 228 distinct RFLP patterns were retrieved, representing 4 distinct patterns for the enzyme *Mlu*I, 99 for enzyme *Hinc*II, and 14 for enzyme *Sac*II (Suppl. Table 1). The analysis of trends of RFLP detection revealed considerable temporal changes (Fig. 2). During 2014, a rapid increase was observed in the number of detections of RFLP 1-7-4,

Table 1. Changes in the detection of restriction-fragment length polymorphism (RFLP) patterns of porcine reproductive and respiratory syndrome virus over time.

RFLP	No. of detections in 2007 from 1,162 sequences	No. of detections in 2019 from 4,265 sequences
Most abundant RFLPs with increased detection over time		
2-5-2	69 (6.1)	919 (21.7)
1-7-4	37 (3.3)	864 (20.4)
1-3-4	19 (1.7)	125 (2.9)
1-3-2	46 (4.1)	382 (9.0)
1-12-4	7 (0.6)	108 (2.5)
Most abundant RFLPs with decreased detection over time		
1-4-2	219 (19.4)	120 (2.8)
1-18-4	113 (10)	0 (0)
1-18-2	61 (5.4)	1 (0.1)
1-2-2	17 (1.5)	8 (0.2)

Numbers in parentheses are percentages.

making it the most prevalent wild-type pattern detected at ISU-VDL (Table 1).

Some notable RFLPs had variable changes in detection over time (Fig. 2). RFLP 1-8-4 had 184 of 1,131 (16.3%) detections in 2007, reached the lowest number in 2013 at 101 of 2,890 (3.5%), and increased to 680 of 4,236 (16.1%) in 2019 (Fig. 2). RFLP 1-4-4 had 88 of 1,131 (7.6%) detections in 2007, reached the highest number in 2012 at 810 of 3,998 (20.3%), and decreased to 363 of 4,236 (8.6%) in 2019. RFLP 1-26-2 was not detected in 2007, reached the highest number in 2013 at 357 of 2,890 (12.4%), and decreased to 10 of 4,236 (0.3%) in 2019. RFLP 1-22-2 was not detected in 2007, reached the highest number in 2012 at 106 of 3,998 (2.6%), and decreased to 1 of 4,236 (0.02%) in 2019. RFLP 1-2-4 had 3 of 1,131 (2.9%) detected in 2007, reached the highest number in 2011 at 128 of 3,196 (4%), and decreased to 34 of 4,236 (0.80%) in 2019 (Fig. 2).

Analysis of the specimens used for the PRRSV ORF5 sequencing revealed a clear trend moving from the usage of individual samples such as serum, to pen-based samples, such as oral fluids (OFs) and processing fluids (PFs). In 2007, serum represented 671 of 1,131 (59.3%) and tissue-lung 419 of 1,131 (37%) specimens. In 2019, serum dropped to 1,197 of 4,236 (28.3%) and tissue-lung to 1,079 of 4,236 (25.5%). Conversely, OFs represented 1,208 of 4,236 (28.5%) and PFs 579 of 4,236 (13.7%) specimens with ORF5 sequencing performed in 2019 (Fig. 3).

Across the 228 distinct RFLP patterns, 4 represented 23,674 of 40,454 (58.5%) ORF5 sequences and were retained for the pairwise and cluster analyses over time and across states, namely: RFLP 2-5-2 (n=8,774; 21.7%), RFLP 1-7-4 (n=6,127; 15.2%), RFLP 1-4-4 (n=4,774; 11.8%), and RFLP 1-8-4 (n=3,999; 9.9%).

Cluster analysis over time

The mean pairwise distances between RFLP 2-5-2 (n=8,774) was 0.69 (SE 0.12). When observing pairwise distances over the years, the largest pairwise distance was revealed for 2007 versus 2008 at 2.1% (SE 0.22) and the shortest for 2011 versus 2012 at 0.6% (SE 0.11; Suppl. Table 2).

For RFLP 2-5-2, the DAPC did not reveal accumulation of relevant genetic changes between 2007 and 2019, forming a single cluster over time (Fig. 4). For RFLP 1-7-4, the DAPC revealed the accumulation of significant genetic changes in 2007–2019. The first principal component of DAPC (Fig. 4, RFLP 1-7-4, horizontal line) presented 2 markedly isolated clusters, in which the first comprised 2007 and 2013, and the second comprising the period after 2014, suggesting that more intense genetic changes were accumulated during 2013 to 2014 than during previous or upcoming years. The second principal component (Fig. 4, RFLP 1-7-4, vertical line) of the DAPC revealed a continuous accumulation of genetic changes across 2014–2019.

The mean pairwise distances for RFLP 1-7-4 over time (n=6,127) was 3.01 (SE 0.23). Within a year, the largest pairwise distances were observed in 2013 at 6.7% (SE 0.58) and the smallest in 2014 at 1.33% (SE 0.16). When observing pairwise distances between subsequent years, the largest pairwise distance was revealed for 2012 versus 2013 at 6.9% (SE 0.55) and the smallest for 2014 versus 2015 at 1.5% (SE 0.16; Suppl. Table 2).

For RFLP 1-8-4, 153 ORF5 sequences were classified as "Prevacent-like" and were removed. The DAPC revealed the accumulation of significant genetic changes across 2007–2019. The second principal component of DAPC (Fig. 4, RFLP 1-8-4, vertical line) revealed more accumulation of genetic changes across 2010 to 2013 compared with 2007 to 2009. The marked change in the first principal component (Fig. 4, RFLP 1-8-4, horizontal line) of the DAPC from 2012 to 2014 revealed the accumulation of genetic diversity across 2012–2013. The second principal component (Fig. 4, RFLP 1-8-4, vertical line) of the DAPC revealed a lower magnitude but continued accumulation of genetic changes across 2014–2019.

The mean pairwise distances for RFLP 1-8-4 over time (n=3,846 sequences) was 9.8% (SE 0.66). Within a year, the

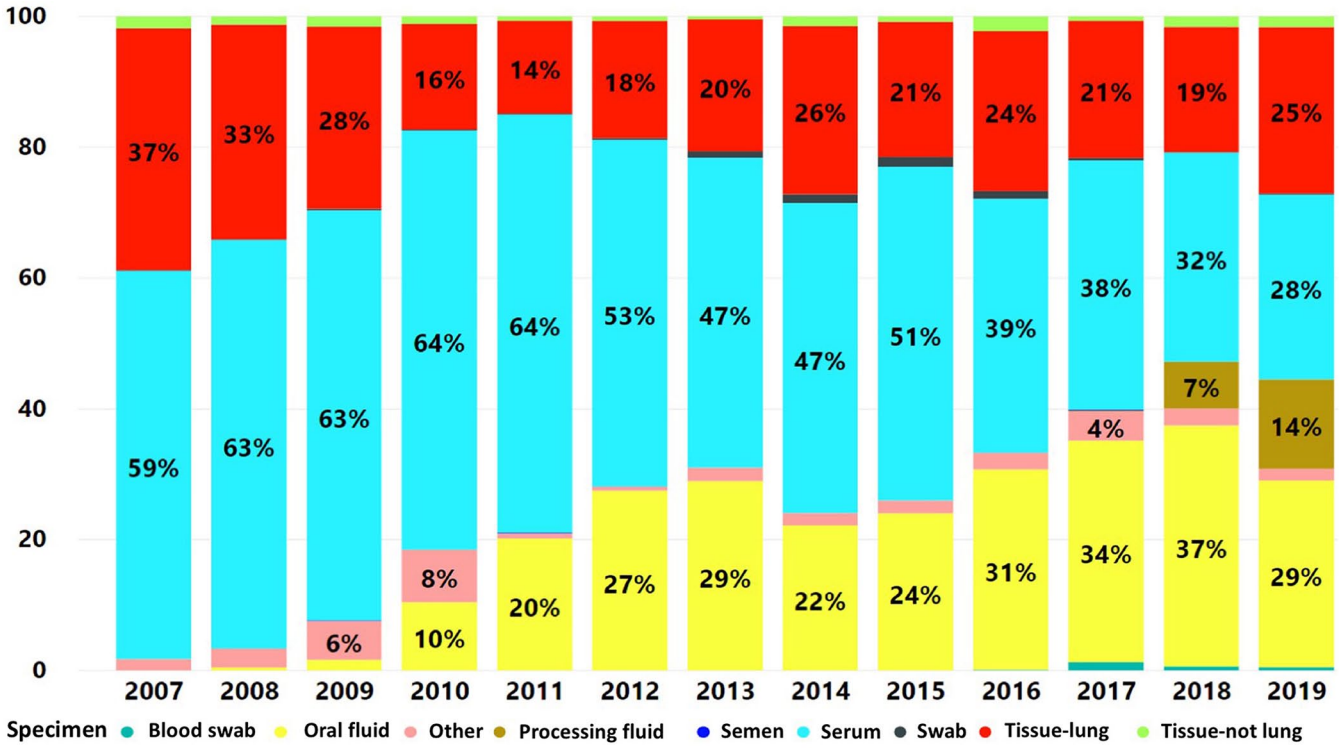

Figure 3. Specimen types submitted for open reading frame 5 (ORF5) sequencing over time. The x-axis labels are the years of detection, 2007–2019. The y-axis represents the percentage of detections within each year.

largest pairwise distances were observed in 2018 at 8.82% (SE 0.64) and the shortest in 2012 at 5.62% (SE 0.43). When observing pairwise distances between subsequent years, the largest pairwise distance was revealed for 2013 versus 2014 at 9.4% (SE 0.72) and the smallest for 2011 versus 2012 at 6.6% (SE 0.52; Suppl. Table 2).

For RFLP 1-4-4, 46 ORF5 sequences were classified as "Primepac-like", and 12 as "Ingelvac ATP–like". Those vaccine-like sequences were removed from the cluster analyses. The change in direction from the second principal component (Fig. 4, RFLP 1-4-4, vertical line) to the first principal component (Fig. 4, RFLP 1-4-4, horizontal line) of the DAPC around 2012–2013 revealed accumulation of significant genetic changes.

The mean pairwise distances for RFLP 1-4-4 over time (n=4,716 sequences) was 8.8% (SE 0.56). Within a year, the largest pairwise distances were observed in 2019 at 10.2% (SE 0.68) and the shortest in 2009 at 5.8% (SE 0.45). When observing pairwise distances between subsequent years, the largest pairwise distance was revealed in 2017 versus 2018 at 10.5% (SE 0.67) and the shortest for 2012 versus 2013 at 6.1% (SE 0.44; Suppl. Table 2).

Cluster analysis across states

For RFLP 2-5-2, 2,436 ORF5 sequences from 14 of 31 U.S. states were retained to conduct DAPC and pairwise comparisons. DAPC analysis did not reveal a significant accumulation of genetic changes across states, forming a single cluster (Fig. 5, RFLP 2-5-2).

The mean pairwise distance was 0.70 (SE 0.13; Fig. 5, Suppl. Table 3). The largest pairwise distances between states were observed between Colorado (CO) versus Indiana (IN) at 1.1% (SE 0.17). The shortest pairwise distance was detected between Utah (UT) versus Kansas (KS) at 0.5% (SE 0.11). Within a state, the pairwise comparison revealed the largest distance for CO at 1.1% (SE 0.20) and the shortest in KS at 0.4% (SE 0.12; Suppl. Table 3).

For RFLP 1-7-4, 2,470 ORF5 sequences from 11 U.S. states were retained for DAPC and pairwise comparisons. The DAPC analysis revealed a significant accumulation of genetic changes across states, forming 4 distinct clusters (Fig. 5, RFLP 1-7-4), consisting of 1 for North Carolina (NC); 1 for Nebraska (NE); 1 including Minnesota (MN), South Dakota (SD), Iowa (IA), Illinois (IL), and Missouri (MO); and 1 including Ohio (OH), IN, Michigan (MI), and Pennsylvania (PA). The mean pairwise distances among states was 3.6% (SE 0.26; Suppl. Table 3). The largest pairwise distance between states was detected between PA versus SD at 4.8% (SE 4.75) and the smallest between NC versus MO at 3.2% (SE 0.28; Suppl. Table 3). The within-state comparison with the largest pairwise distance was detected in SD at 4.7% (SE 0.44) and the smallest in NE at 2.8% (SE 0.29; Suppl. Table 3). There was a mean genetic

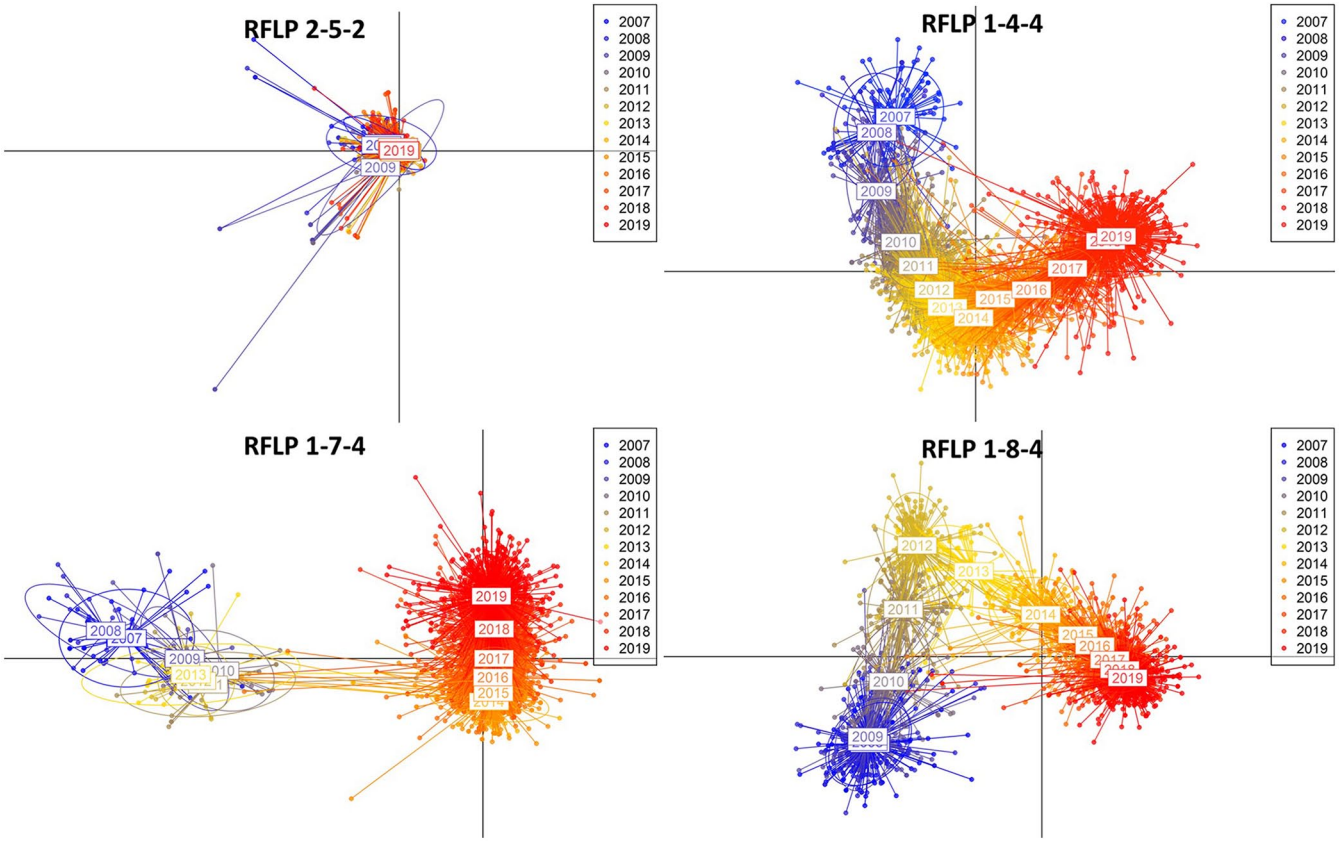

Figure 4. Discriminant analysis of principal components for different restriction-fragment length polymorphism (RFLP) for 2007–2019. Groups sharing genetic similarities are represented by ellipses, each state is represented by a different color, and each dot represents an individual porcine reproductive and respiratory syndrome virus open reading frame 5 (ORF5) sequence.

distance of 3.6% between U.S. states for 2017 to 2019, and the unidirectionality of overlapping clusters for RFLP 1-7-4 (Fig. 4, RFLP 1-7-4) provides evidence of expanding genetic diversity of PRRSV strains classified within RFLP 1-7-4 during recent years (Fig. 5, RFLP 1-7-4).

For RFLP 1-8-4, 1,657 ORF5 sequences from 10 states were retained for DAPC and pairwise comparisons. DAPC analysis revealed a significant accumulation of genetic changes across states, forming 4 distinct clusters (Fig. 5, RFLP 1-8-4), composed of 1 for MO; 1 for Oklahoma (OK); 1 for OH and IN; and 1 for IA, IL, MN, NE, KS, and CO. The principal component of DAPC (Fig. 5, RFLP 1-8-4, horizontal line) revealed more intense accumulation of genetic changes from MO compared to all other states. The second component of DAPC (Fig. 5, RFLP 1-8-4, vertical line) revealed a high accumulation of genetic changes across the other states. The mean pairwise distances among states were 8.7% (SE 0.60; Suppl. Table 3). The largest pairwise distance was revealed between OK versus MO at 13.3% (SE 0.51) and the shortest between KS versus OH at 4.5% (SE 0.53). The within-state comparison revealed the largest pairwise distance for IA at 7.2% (SE 0.53) and the shortest for OH at 2.1% (SE 0.23; Suppl. Table 3).

For RFLP 1-4-4, 754 ORF5 sequences from 9 states were retained for DAPC and pairwise comparisons. DAPC analysis revealed a significant accumulation of genetic changes across states, forming 5 distinct clusters (Fig. 5, RFLP 1-4-4): 1 for OK; 1 for NC; 1 for NE; 1 for IN; and 1 for OH, MN, MO, IL, and IA. The principal component of DAPC (Fig. 5, RFLP 1-4-4, horizontal line) revealed more accumulation of genetic changes across OK compared to all other states. The second component of the DAPC (Fig. 5, RFLP 1-4-4, vertical line) revealed most accumulation of genetic changes across the other states. The mean pairwise distances among states were 10.2% (SE 0.66; Suppl. Table 3). The largest pairwise distance between states was detected between OK versus NC at 14.5% (SE 1.15) and the shortest between MO versus NC at 6.2% (SE 0.47). The within-state comparison with the largest pairwise distance was detected in IN 10.5% (SE 0.68) and the shortest in MO 3.6% (SE 0.33).

Lineage comparison

All sequences from RFLP 2-5-2 were grouped within reference sequences of lineage 5 and with the original referent

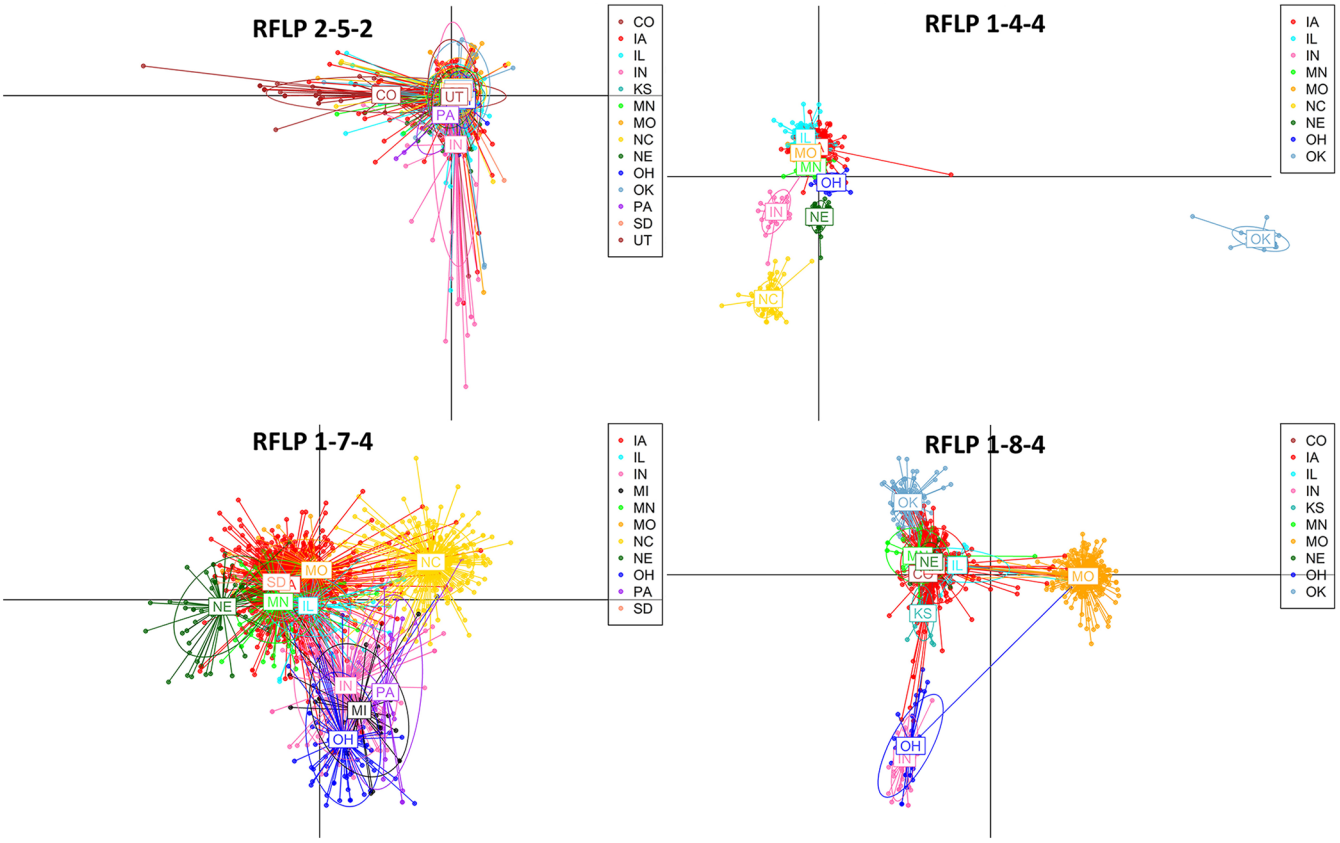

Figure 5. Discriminant analysis of principal components across states for different restriction-fragment length polymorphism (RFLP) for 2017–2019. Only states with at least 10 open reading frame 5 (ORF5) sequences recovered by RFLP pattern were included in the analysis. Groups sharing genetic similarities are represented by ellipses, each state is represented by a different color, and each dot represents an individual porcine reproductive and respiratory syndrome virus ORF5 sequence.

sequence ATCC VR-2332 (GenBank U87392). All PRRSV ORF5 sequences classified as RFLP 1-7-4 and RFLP 1-8-4 grouped with reference sequences from lineage 1. RFLP 1-4-4 sequences were grouped mostly (76.9%, $n=3{,}628$) within reference sequences of lineage 1C, and additionally with reference sequences from lineages 1A, 5, 7–9 (21.5%, $n=1{,}013$; 0.5%, $n=24$; 0.1%, $n=4$; 0.5%, $n=23$; 0.5%, $n=24$, respectively).

Discussion

We encountered 228 distinct RFLP patterns from samples submitted for testing on a fee-for-service basis at the ISU-VDL. To our knowledge, this is the largest number of RFLP patterns reported in a single project; 117 RFLP patterns were reported in a single dataset in a Canadian study.[22] The ever-increasing genetic diversity of PRRSV could be demonstrated by the increasing number of different RFLP patterns detected for each of the enzymes *Mlu*I, *Hinc*II, and *Sac*II. A 1998 work reported 2 different patterns for *Mlu*I.[66] In 2010, 3 were reported,[33] and in the present work, we detected 4. For the enzyme *Hinc*II, there were 4 in 1998,[66] 40 in 2010,[33] and 99 in our work here. For the enzyme *Sac*II, there were 4 in 1998,[66] 8 in 2010,[33] and 14 in our work here.

We detected RFLP 2-5-2 most frequently, accounting for 21.2% of all ORF5 sequences. The RFLP 2-5-2 sequences formed only one cluster with a mean pairwise distance of 0.7% across 8,774 ORF5 sequences, demonstrating that RFLP 2-5-2 remains genetically conserved. The low amount of genetic diversity encountered for RFLP 2-5-2 may occur as a consequence of swine exposure to a vaccine-like strain, and each time a pig is vaccinated with the respective PRRS MLV vaccine, a new exposure to the referent vaccine strain occurs. The Sanger[52] method employed for ORF5 sequencing reports a single ORF5 consensus sequence; an alternative sequencing method facilitates the recovery of a wild-type strain when a vaccine-like and a wild-type strain are present in the same specimen.[9] The increased detection of RFLP 2-5-2 after 2010 that we detected here is aligned with an increased proportion of U.S. breeding herds using PRRS MLV vaccines.[41,53,58]

RFLP 1-7-4 has been detected since 2007 and had relatively low detection until 2013. We observed important genetic changes in 2012–2013 in RFLP 1-7-4 that may have been the result of the appearance of a more pathogenic strain causing significant disease and an increased need for testing and sequencing, resulting in recovery of a very similar strain during 2014 with an overall pairwise distance of 1.3% for

that year. The first reported field complaints related to RFLP 1-7-4 being associated with clinical cases occurred in NC during 2014.[31] During that time of increased detection of RFLP 1-7-4, there were simultaneous reports of high production losses in infected herds.[1,24,25,40] In the original work that proposed the use of RFLP patterns to differentiate vaccine-like strains from the wild-type strain, RFLP 1-7-4 was not present.[66] A retrospective study estimated that this virus strain appeared around 1999 and was circulating in sow herds until 2014 without significant changes in its genetic diversity.[1] Our finding here of the genetic change in RFLP 1-7-4 agrees with the changes reported in previous Bayesian statistical methods[1] and recombinant analysis.[62]

The second wild-type family detected most frequently was RFLP 1-8-4, which was detected since the beginning of the study in 2007. Historically, RFLP 1-8-4 caused tremendous production losses and economic impact in MN during the early 2000s,[8] and was the pattern detected most frequently at the University of Minnesota Veterinary Diagnostic Laboratory (UMN-VDL; St. Paul, MN, USA) during 2004–2007.[31] Introduction of RFLP 1-8-4 into the United States was attributed to infected feeder pigs from Canada given that the strain classified as RFLP 1-8-4 detected in the United States had high genetic similarity with Canadian strains.[8] RFLP 1-7-4 is the most frequent wild-type strain detected in the United States; RFLP 1-8-4 remains the most frequent wild-type strain detected in Canada.[22,23]

Using cluster analysis, we detected genetic change in RFLP 1-8-4 from 2010 to 2014, which coincided with decreased detections of RFLP 1-8-4. After 2014, increased detection of RFLP 1-8-4 was observed, which coincided with the most significant accumulation of genetic changes for 2012–2014. RFLP 1-8-4 formed 4 distinct clusters during 2017 to 2019 and had lower within-state pairwise distances compared with between-states, which demonstrates that the RFLP 1-8-4 family is still evolving with increasing geographic diversity. This finding suggests that the RFLP 1-8-4 is more conserved within states, such as OK and MO, than between those states. These findings support the hypothesis of genetic variability across U.S. states for strains sharing the same RFLP pattern.

The third most frequently detected wild-type was RFLP 1-4-4. Different than RFLP 1-7-4 that had increased detection during recent years, RFLP 1-4-4 was detected most commonly until 2014. The cluster analysis for RFLP 1-4-4 demonstrated a consistent evolutionary pattern over time with an accumulation of genetic changes during 2012–2013. RFLP 1-4-4 formed 5 distinct clusters between states, with the most considerable mean pairwise distance observed between OK versus NC at 14.5%. This finding suggests that RFLP 1-4-4 is more diverse across states. The large diversity within RFLP 1-4-4 strains was also supported in our work by the detection of this strain within the different genetic lineages 1, 5, 7–9.

During 2017–2019, there was an unexpected formation of distinct clusters between U.S. states for RFLPs 1-7-4, 1-8-4, and 1-4-4. The differences encountered for U.S. states could be potentially linked with the outflow of grower pigs from some states to other regions and inflows, especially of replacement animals, with most of them naïve for PRRSV. The current U.S. swine flow and production characteristics could play a significant role in maintaining and increasing geographic isolation of the virus. This hypothesis is supported by the fact that NC is a typical provider of feeder pigs to the Midwest, and this pig movement has been associated with the introduction of new strains of PRRSV to the Midwest.[33,55] NC is the state with the second-largest hog inventory in the United States,[34] and had only 10 RFLP 1-8-4 detections from 2007–2019. On the other hand, the inflow of PRRSV naïve replacement animals to NC breeding herds may have kept some of the notable strains out of the NC herds.

Some RFLPs, such as 1-26-3, 1-22-2, 1-18-2, and 1-4-4, were detected more frequently in some periods of time. Virulence characteristics of these strains may have influenced this trend of detection. It is known that some strains classified within RFLP 1-4-4 have moderate virulence[4] compared with highly virulent strains classified within RFLP 1-7-4.[62] Even though there is no clear predictor of disease severity in the PRRSV genome, despite the encountered RFLP pattern, there were no other genetic domains present in the highly pathogenic strains that were not present in mildly pathogenic strains.[62] In contrast, mild strains can survive longer in the field given the lower impact on the productivity of these strains.

The wild-type strain for RFLP 1-3-2 was detected in our study since 2007. Recent changes with increased detection of RFLP 1-3-2 is the result of detecting "Fostera-like" vaccine strains contributed by the new Fostera PRRSV MLV vaccine, which was introduced in the international market in 2012.[39]

Even though RFLP designations have their limitations, until a better tool to classify PRRSV strains is developed, or understood by practitioners, such as the newer genetic lineage classification,[37,55] the RFLP pattern will be available and can be used as a screening approach to inform changes in the trends of PRRSV detection over time and select ORF5 sequences for cluster analysis. It is known that an increase in genetic diversity in a RNA virus is mainly caused by poor fidelity in the replication process.[3] Furthermore, an evolutionary investigation by phylogenetic analysis provides more robust epidemiologic information than other methods.[33] Additional phylogenetic comparisons between recently sequenced strains and the historical PRRSV accumulated in a database is still highly encouraged given that PRRSV mutates over time.[3,33]

It is crucial to understand genetic variability within RFLPs (up to 14.5% for 1-4-4s) to develop PRRSV regional control or elimination programs,[6,30] herd immunization strategies, and support adoption of adequate biosecurity practices.[56] Additionally, regional differences in PRRSV genetic diversity may help design appropriate animal movements across state borders and regional placement strategies to minimize the risk of cross-contamination by different PRRSV strains.

The specimens used for ORF5 sequencing have changed considerably over time. In 2007, serum and lung (tissue-lung) were the primary specimens used for ORF5 sequencing. In contrast, after the description, validation, and adoption of new pen-based sampling methods for PRRSV testing based on OF[42–44,46,51] and PF,[26,27,59,61,63,64] OFs and PFs were also used for ORF5 sequencing, in agreement with other reports.[60]

Limitations of our study include the use of only ORF5 sequences to compare genetic variability across PRRSV strains. A PRRSV whole-genome sequencing (WGS) comparison could be more revealing, but it is currently limited because of the lower number of PRRSV WGS sequences available. Inclusion of data from other VDLs beyond the ISU data could potentially reveal more characteristics of the current genetic variability encountered for PRRSV in the United States. Additionally, we did not investigate genetic diversity between different RFLP patterns.

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Availability of data and materials

Charts that report RFLP pattern detection over time were transferred to the Swine Disease Reporting System (SDRS), are available online at the project website www.fieldepi.org/SDRS, and are regularly updated with data from ISU-VDL and other SDRS participant VDLs. VDL confidentiality restricts the public availability of the PRRSV ORF5 sequences.

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Supplemental material

Supplemental material for this article is available online.

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